

Q1. You are a movie enthusiast whose interest changes depending on the season. During summer, you get addicted to Action Movies. Then winter comes and you start watching Romantic Movies. Then festive season comes, you spend most of your time watching Comedy Movies with friends. Then in the Rainy Season, you again start watching Romantic movies. Towards the end of rainy, you switch to Action movies. Then major award season (Oscars) comes up, and your focus shifts to Documentary Movies. However, after the buzz ends, you return to Comedy Movies.

Task:

Draw the state diagram of your movie preference changes throughout the year and A Circuit Diagram using T Flip-Flops. Show all states and transitions.

[For any unused states/transitions use Don't Care]

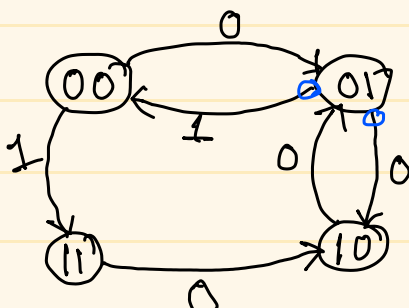
Step 1: Find total States $\rightarrow$ Action (A)
Romantic (R)
Comedy (C)
Documentary (D)

4 states
 $\lceil \log_2 4 \rceil = 2$ FF

Step 2: Make normal SD : $A \rightarrow R \rightarrow C \rightarrow R \rightarrow A \rightarrow D \rightarrow C$
00 01 10 01 00 11 10

Step 3: Make Diagram and find how many external inputs needed

First outgoing edge from a state is 0, then 1, then 2, ...



if the max num of outgoing edge from any state is m then the no of external variable $\lceil \log_2 m \rceil = \lceil \log_2 3 \rceil = 2$

Step 4: Make State Table : [use ET to Find FF inps]

3 outgoing edge
 $\lceil \log_2 3 \rceil = 2$ ext inp

Present State			External Input		Next State		Flip Flop	
A	B	x	A^+	B^+	T_A	T_B		
0	0	0	0	1	0	1		
0	0	1	1	1	1	1		
0	1	0	1	0	1	1		
0	1	1	0	0	0	1		
1	0	0	0	1	1	1		
1	0	1	X	X	X	X		
1	1	0	1	0	0	1		
1	1	1	X	X	X	X		

ET		
Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

FF inputs
TA, TB using Kmap

★ STEP 5: Find

TA:

	Bx	00	01	11	10
A					
0		0	1	0	1
1		1	x	x	0

$$TA = A\bar{B} + \bar{B}x + \bar{A}B\bar{x}$$

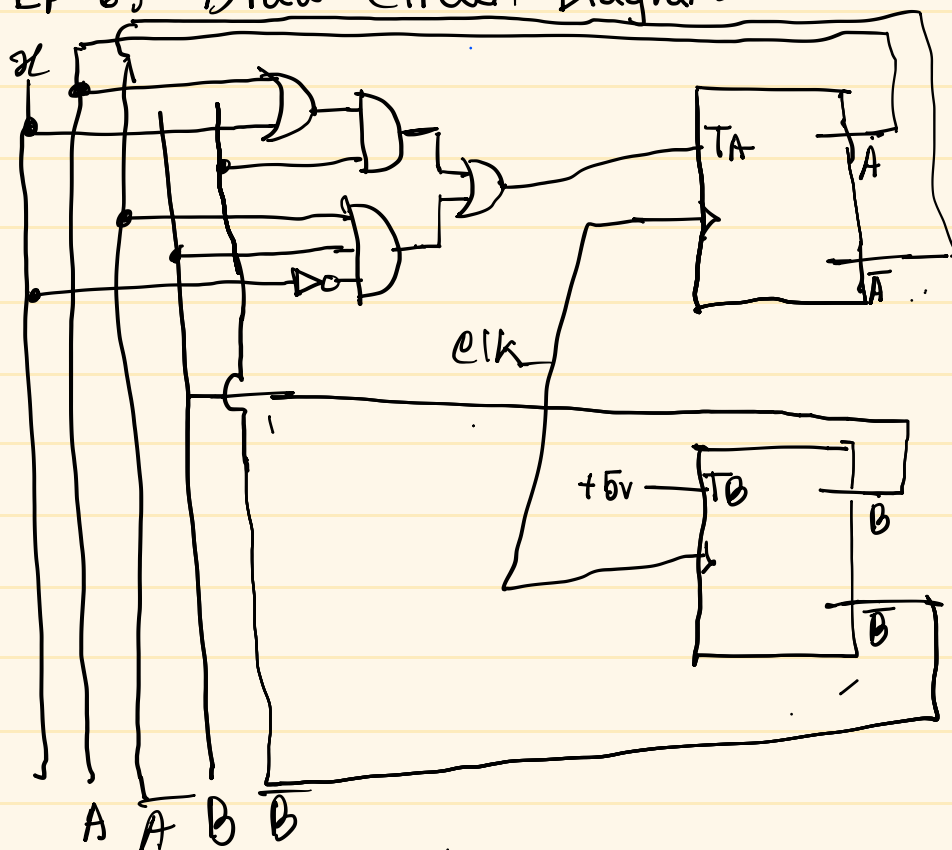
$$= \bar{B}(A+x) + \bar{A}B\bar{x}$$

TB:

	Bx	00	01	11	10
A					
0		1	1	1	1
1		1	x	x	1

$$TB = 1$$

STEP 6: Draw Circuit Diagram



For Practice Prob-3 see below

[use X for unused states / transitions]

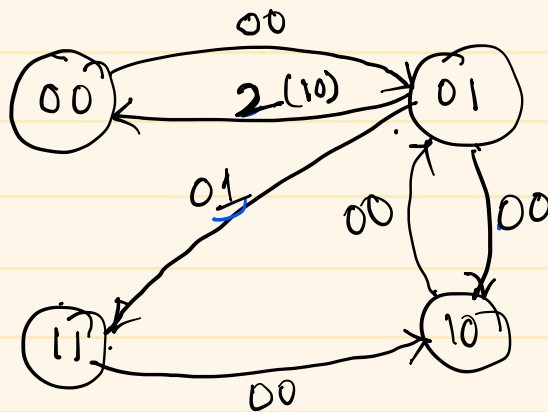
Practice Problem 3 From slide:

Green₀₀ → Yellow₀₁ → Red₁₀ → Yellow₀₁ → Blue₁₁ → Red₁₀

→ Yellow₀₁ → green₀₀

4 states → 2 FF →

G 00
Y 01
R 10
B 11



3 outgoing edges from yellow,
so no of external
inp = $\lceil \log_2 3 \rceil = 2$
(w₂x)

ST₀

PS		E inp		NS		FF	
A	B	w	x	A+	B+	DA	DB
0	0	0	0	0	1	0	1
0	0	0	1	X	X	X	X
0	0	1	0	X	X	X	X
0	0	1	1	X	X	X	X
0	1	0	0	1	0	1	0
0	1	0	1	1	1	1	1
0	1	1	0	0	0	0	0
0	1	1	1	X	X	X	X
1	0	0	0	0	1	0	1
1	0	0	1	X	X	X	X
1	0	1	0	X	X	X	X
1	0	1	1	X	X	X	X
1	1	0	0	1	0	1	0
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

★ DA

AB \ wx	00	01	11	10
00		X	X	X
01	1	1	X	
11		X	X	X
10	1	X	X	X

$$D_A = A\bar{B} + \bar{A}B\bar{0}$$

DB

AB \ wx	00	01	11	10
00	1	X	X	X
01		1	X	
11	1	X	X	X
10		X	X	X

$$D_B = 1 + AB + \bar{A}\bar{B}$$

$$= 1 + (A \odot B)$$

